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Windows Forensics

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Computer Forensics

Windows Registry

Registry: Key Terms

The Windows Registry is organized in a specific way that has changed slightly with each new version of Windows.

- **Registry:** A collection of files containing system and user information.

- **Registry Editor:** A Windows utility for viewing and modifying data in the Registry.

- There are two Registry Editors:
  - Regedit and
  - Regedt32 (introduced in Windows 2000).

Registry: Key Terms

- **HKEY:** Windows splits the Registry into categories with the prefix HKEY_.
- Windows 9x systems have six HKEY categories and Windows 2000 and later have five.
- Windows programmers refer to the "H" as the handle for the key.
- **Key:** Each HKEY contains folders referred to as keys. Keys can contain other key folders or values.

Registry: Key Terms

- **Subkey:** A key displayed under another key is a subkey, similar to a subfolder in Windows Explorer.
- **Branch:** A key and its contents, including subkeys, make up a branch in the Registry.
- **Value:** A name and value in a key; it's similar to a file and its data content.
- **Default value:** All keys have a default value that may or may not contain data.

Registry: Key Terms

- **Hives:** Hives are specific branches in HKEY_USER and HKEY_LOCAL_MACHINE.
- Hive branches in HKEY_LOCAL_MACHINE\Software are SAM, Security, Components, and System.
- For HKEY_USER, each user account has its own hive link to Ntuser.dat.

Registry: Overview

- According to Microsoft, the Windows Registry in a "Central Hierarchical Database" designed "to store information that is necessary to configure the system for one or more users, applications and hardware devices."
- The Windows Registry replaced the configuration and initialization (ini) files used in Windows versions prior to Windows NT.
- In addition, the Registry records information specific to users, essentially tracking a user's activity in order to organize and optimize the user's experience while interacting with the system.

Registry: Overview

- Files that comprise the Registry and contain information about the system and applications are located in the Windows\system32\config directory.

- Additional files that contain user-specific information and application settings are found in the user profile directories (i.e., NTUSER.DAT and USRCLASS.DAT).

Registry: Overview

- Most Important Files for Registry Analysis are:
  - NTUSER.DAT
  - USERCLASS.DATA
  - Software
  - System
  - SAM
  - Security

- Each registry file begins with the header regf, and each entry starts with a header hbin. The hbin entries can be unallocated (or deleted) and recovered manually or with carving tools.

Registry: Overview

Filename and location	Description
Windows 9x/Me	
Windows\System.dat	User-protected storage area; contains installed program settings, usernames and passwords associated with installed programs, and system settings
Windows\User.dat Windows\profile\UserAccount	Contains the most recently used (MRU) files list and desktop configuration settings; every user account created on the system has its own user data file
Windows NT, 2000, XP, and Vista	
Documents and Settings\user-account\ Ntuser.dat (in Vista, Users\UserAccount\Ntuser.dat)	User-protected storage area; contains the MRU files list and desktop configuration settings
Winnt\system32\config\Default	Contains the computer's system settings
Winnt\system32\config\SAM	Contains user account management and security settings
Winnt\system32\config\Security	Contains the computer's security settings
Winnt\system32\config\Software	Contains installed programs settings and associated usernames and passwords
Winnt\system32\config\System	Contains additional computer system settings

Registry: SOFTWARE

- SOFTWARE file exists on disk at %systemroot%\Windows\system32\config.

- The software file contains information on installed software, including the Windows OS and Microsoft applications. Any application can store configuration data in the software key.

- With the addition of 64-bit OSs, Microsoft created a method to allow compatibility with 32-bit applications.

- These separate entries exist in the software key under HKey\LocalMachine\Software\WOW6432Node.

Registry: SYSTEM

- SYSTEM file exists on disk at %systemroot%\Windows\system32\config. When a system is running, a current control set is loaded into the software hive.

- The key itself contains two or more control sets – the last known good and the current control sets –in the event the registry becomes corrupted.

- If examining a dead image, both(or more)control sets should be looked at with the understanding that the current control set was in use the last time the system was booted.

Registry: SAM

- SAM (Security Accounts Manager) file exists on disk at %systemroot%\Windows\system32\config.
- This file contains account information for the local system.
- It includes local users, the user's encrypted password, and the location of the user's registry hive.
- Domain accounts will be stored in the active directory, and may not be stored locally in the SAM file.

Registry: SECURITY

- Contains the security information stored in the key HKLM\SECURITY.

- It contains the security database of the domain info into which the current user is logged in.

- This key is linked to the local machine and the kernel reads it to enforce the security policy applicable to the current user and all applications or operations executed by the user.

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Registry: NTUSER.DAT

- NTUSER.DAT file contains information specific to a user, including settings, most recently used (MRU) keys, start-up scripts or applications, preferences, and more. Each user with a local profile on a system has an NTUSER.DAT file.

- The file is located by default in C:\Users\{profile_name}\ for Windows Vista and later C:\DocumentsandSettings\{profile_name}.

- Data contained in the NTUSER.DAT can indicate files the user accessed, URLs typed into a browser or Explorer bar, commands the user-initiated, software the user installed, and file shares the user mapped.

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Registry: USERCLASS.DAT

- USERCLASS.DAT File was added in Windows 7. It is located by default in
- “C:\Users\{profile_name}\AppData\Local\Microsoft\Windows\”
- This file contains some information previously stored in the NTUSER.DAT. Specifically, the Shell-Bag information for a user profile is now stored here.
- ShellBags are registry artifacts that store user preferences for directories such as window size, sorting order, and placement on the screen.

Registry: Hives

- Registry hive files are made up of 4-KB sections or “bins.”
- These bins are meant to make the allocation of new space (as the hive file grows), as well as the maintenance of the hive file itself, easier. The first 4 bytes of a normal hive file start with “regf” (or 0x66676572).
    - HKEY_CLASSES_ROOT (HKCR)
    - HKEY_CURRENT_USER (HKCU)
    - HKEY_LOCAL_MACHINE (HKLM)
    - HKEY_USERS (HKU)
    - HKEY_CURRENT_CONFIG (HKCC)

Registry: Hives

- The first hbin marker is very important, as this is the base location for offset values listed with the key and value cells throughout the rest of the hive file.

HKEY_CLASSES_ROOT (HKCR)

- Contains filename extension associations like .exe. Also contained in this hive are COM objects, Visual Basic programs, and other automation.

- The Component Object Model (COM) allows nonprogrammers to write scripts for managing Windows operating systems.

HKEY_CURRENT_USER (HKCU)

- Stores configuration information (related to the installed software and operating system) to the currently logged-in user.
- This profile changes each time the user logs in to the system.
- A user profile includes desktop settings, network connections, printers, and personal groups.
- This hive contains very little data but acts as a pointer to HKEY_USERS.

HKEY_LOCAL_MACHINE (HKLM)

- Contains information about the system's settings, including information about the computer's hardware and operating system.

- Contains the majority of the configuration information for currently installed programs and the Windows OS itself.

HKEY_USERS (HKU)

- Contains information about all of the registered users on a system.

- Within this hive are a minimum of three keys.
  - First key is .DEFAULT, which contains a profile when no users are logged in.

  - Second key containing the SID for the current local user, which could look something like S-1-5-18.

  - Third key for the current user with _Classes at the end. For example, HKEY_USER S\S-1-5-21-3794263289-4294853377-1685327589-1003_Classes.

HKEY_CURRENT_CONFIG (HKCC)

- Does not store information itself; instead, acts as a pointer to another registry key
  - (HKEY_LOCAL_MACHINE\System\CurrentControlSet\Hardware Profiles\Current).

- This hive keeps the information about the hardware profile of the local computer system.

- Contains information pertaining to the system's hardware that is necessary during the startup process.

- Within this hive are screen settings, screen resolution, and fonts. Information about the plug-and-play BIOS is also found in this hive.

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Auto Startup Locations

- Windows has a feature that allows programs to launch automatically as it boots; this feature is necessary for some applications like antivirus software that must run first to stop any malicious software before Windows gets booted completely.
- Malicious software like keyloggers and botnets can add entries to the Windows registry in order to launch automatically with each Windows boot.
- The Windows registry stores a record of every program booted with Windows.

Registry: Overview

- Not all applications create a presence in the Registry. For example, some peer-to-peer (P2P) sharing applications are cross-platform and Java-based, and as such don't rely on the Windows Registry to store information.
- Instead, they use configuration files in order to make cross-platform coding easier.

Registry: Way Forward

- Portions of the Windows Registry visible through the Registry Editor are "volatile," meaning that they are populated when the system is booted or when a user logs in and do not exist on disk when the system is shut off.

Registry: Example

- One example of volatile data is the HKEY_CURRENT_USER hive.
- When viewed through the Registry Editor, you can clearly see this hive, and after a little exploration, you'll find that the information in this portion of the Registry pertains specifically to the logged-on user.
- However, when you shut the system down and analyze an acquired image, you won't find an HKEY_CURRENT_USER hive or any file by that name.
- That's because this hive is populated by using the hive for the user that's logged into the system.

Registry: Example

- For the currently logged-in user, the HKEY_CURRENT_USER\SessionInformation key contains a value named ProgramCount that keeps track of the number of programs you have running on your desktop.
- This is the count you see when you lock your workstation.
- However, this value doesn't exist in the user's NTUSER.DAT file when the system is shut down.

Registry: Importance

- The Registry is perhaps the most overlooked treasure trove of forensic artifacts on a Windows system.
- A great source of evidence and supporting evidence.
- Useful for validating findings through an investigation.

Registry: Structure

- Layout of the Registry
  - Hives
  - Keys
  - Subkeys
  - Values
  - Data
- Various Registry Values types
  - Binary
  - String
  - Hex
  - BLOB

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Registry Key: Overview

- The Registry's binary structure consists of "cells," the two most important of which are keys and values. The data is structured in a tree format. Each node in the tree is called a key. Each key can contain both subkeys and data entries called values.

- Sometimes, the presence of a key is all the data that an application requires; other times, an application opens a key and uses the values associated with the key.

- A key can have any number of values, and the values can be in any form.

Registry: Overview

- Each key has a name consisting of one or more printable characters.
- Key names are not case sensitive. Key names cannot include the backslash character (\), but any other printable character can be used.
- Value names and data can include the backslash character.
- The name of each subkey is unique with respect to the key that is immediately above it in the hierarchy.
- Key names are not localized into other languages, although values may be.

Registry Key: Overview

- the structure of a Registry key starts with a 4-byte DWORD (double word) that comprises the size of the key, which when read as a signed integer produces a negative value.
- followed by the key node identifier “nk,” which tells us that this is a key node or cell.
- after the node identifier is a 2-byte value that describes the node type;
  - 0x2C, which identifies a root key cell.
  - 0x20, which identifies a regular key cell.

Registry Key: Overview

- The key structure also contains a LastWrite time, a 64-bit FILETIME object that marks the number of 100-nanosecond epochs since midnight of 1 January 1601.

- The FILETIME object can be converted to a human-readable date and time format by subtracting the hex value 0x146BF33E42C000

Registry Key Cell Structure

Offset	Size	Description
0	4	Size
4	2	Node ID (“nk,” or 0x6B6E)
6	2	Node Type (0x2C or 0x20)
8	8	LastWrite time
20	4	Offset to this key’s parent key
24	4	Number of subkeys
32	4	Offset to the list of subkey records
36	4	Number of values
44	4	Offset to the value list
48	4	Offset to security identifier record
76	2	Length of the key name

Registry Value Cell Structure

Offset	Size	Description
0	4	Size (as a negative number)
4	2	Node ID (“vk,” or 0x6B76)
6	2	Value name length
8	4	Data length
12	4	Offset to data
16	4	Value type

Registry Value Types

Type	Name	Description
0	REG_NONE	No value type
1	REG_SZ	Unicode null-terminated string; can be Unicode or ASCII
2	REG_EXPAND_SZ	Unicode null-terminated string with environment variables/references
3	REG_BINARY	Binary data (no set length or structure)
4	REG_DWORD	32-bit number
5	REG_DWORD_BIG_ENDIAN	32-bit number
6	REG_LINK	Unicode symbolic link
7	REG_MULTI_SZ	Multiple Unicode strings, each "\00" terminated
8	REG_RESOURCE_LIST	Resource list (resource map)
9	REG_FULL_RESOURCE_DESCRIPTOR	Resource list (hardware description)
10	REG_RESOURCE_REQUIREMENTS_LIST	A series of nested arrays that store information about device drivers
11	REG_QWORD	64-bit number

Registry Key Locations

Registry Key Locations
HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Explorer\ShellServiceObjects
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\RunServicesOnce
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\RunOnce
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer\Run
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Run
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows NT\CurrentVersion\Windows
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\ShellServiceObjectDelayLoad
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Explorer\SharedTaskScheduler
HKEY_LOCAL_MACHINE\Software\Microsoft\Active Setup\Installed Components
HKEY_LOCAL_MACHINE\Wow6432Node\Microsoft\Active Setup\Installed Components

Registry Key Locations

Registry Key Locations
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Explorer\SharedTaskScheduler
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows NT\CurrentVersion\Drivers32
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\RunOnce
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer\Run
HKEY_CURRENT_USER\Software\Microsoft\Windows NT\CurrentVersion\Windows\load
HKEY_CURRENT_USER\SOFTWARE\Wow6432Node\Microsoft\Windows\CurrentVersion\Run (64 bit systems only)
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\RunServices
HKEY_CURRENT_USER\ Software\Microsoft\Windows\CurrentVersion\RunServicesOnce
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\RunOnceEx

Computer Forensics

Event Logs

Event Log

- Windows event logs:
  - System Logs
  - Security Logs
  - Application Logs
  - Setup
  - Forwarded Events

Event Log

- **Application Logs: These** contain information logged by applications on the system.
- Security Logs: These contain incidents related to security events according to the auditing policy of the Windows operating system. This log contains login attempts (success and failure), elevated privileges, and more. Details of the security event logs can be increased by applying a specialized auditing policy either locally or by a GPO.
- **System Logs:** These contain messages generated by the Windows operating system.

Event Log

- **Setup:** captures incidents of installation or upgrading of the Windows operating system.

- **Forwarded Events:** contains events that are forwarded by other computers. This event is populated if Windows Event Forwarding is enabled and the local machine is running as a central subscriber, other machines are forwarders.

Event Log

- All logs are stored in five categories/levels (information, warning, error, critical, and success/failure audit).

- Event logs are stored in XML format in System32/winevt/Logs folder. On the other hand, the Computer\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\EventLog registry key provides default and customized event log settings.

Event Log

- The artifact contains Event Logs in Windows operating systems:
  - Level - Event log level/type. This can be information, warning, error, or success/failure audit.
  - Channel - Event log channel or category. Security, Application, System, etc.
  - Computer - Local system name.
  - Event ID - Event identification number. By filtering according to ID we can get the important events.
  - Keywords - They are used to group the event with other similar events based on the usage of the events.
  - Opcode - The activity or a point within an activity the application performed when it raised the event.
  - Provider GUID - The unique GUID for the provider. It is useful when performing research or operations on a specific provider.

Event Log

- The artifact contains Event Logs in Windows operating systems:
  - Provider Name - Name of event provider.
  - Security User ID - It uniquely identifies a security principal or security group.
  - Task - Identifies the type of recorded event log. Application developers can define custom task categories for providing additional details.
  - Event Record Order - Order of the event in the main event category.
  - Located At - File offset location of the specific event.
  - Event Record ID - Event record identification number in the main category.
  - XML - XML view of the event,
  - Record Length - Length of the event.

Event Logs: Header

Offset	Size	Field	Description
0x00	4	Length	The length of the entire entry
0x04	4	Reserved	The "LfLe" signature
0x08	4	RecordNumber	The event record number
0x0C	4	TimeGenerated	Time the entry was submitted
0x10	4	TimeWritten	Time the entry was written to the log
0x14	4	EventID	Packed bytes
0x18	2	EventType	Event type (error, failure, success, information, or warning)
0x1A	2	NumStrings	The number of strings in the log entry description
0x1C	2	EventCategory	category of the event specific to the source
0x1E	2	ReservedFlags	Reserved

Event Logs: Data

Variable String	Field	Description
0x00	Length	The length of the entire entry
0x04	Reserved	The “LfLe” signature
0x08	RecordNumber	The event record number
0x0C	TimeGenerated	Time the entry was submitted
0x10	TimeWritten	Time the entry was written to the log
0x14	EventID	Packed bytes
0x18	EventType	Event type (error, failure, success, information, or warning)
0x1A	NumStrings	The number of strings in the log entry description
0x1C	EventCategory	category of the event specific to the source
0x1E	ReservedFlags	Reserved

Event Logs: Header

Offset	Size	Field	Description
0x20	4	ClosingRecordNumber	Reserved
0x24	4	StringOffset	Offset to the description of the log entry
0x28	4	UserSidLength	The size of the UserSID (zero if no user identifier) ($E)
0x2C	4	UserSidOffset	(Variable length $B) Offset to the UserSID
0x30	4	DataLength	Time the entry was submitted
0x34	4	DataOffset	Time the entry was written to the log

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